



Kamalakar Peta

Staff Data Engineer — Modern Data Stack (Databricks · dbt · Airflow)

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Summary

Staff-level Data Engineer with 11+ years building and owning large-scale data pipelines end-to-end across four global financial-data firms. Deep hands-on expertise in the modern data stack — Databricks, dbt, Airflow, PySpark, and SQL — running at 15+ TB/day, with a track record of leading migrations, improving pipeline reliability and cost, and setting data-quality and CI/CD standards for the team. Turns ambiguous requirements into dependable systems, collaborates across analytics and data-science teams, and mentors engineers.

Technical Skills

- **Languages & Query:** Python · SQL · PySpark
- **Data Platform:** Databricks (Unity Catalog, Delta Lake, Workflows, Asset Bundles) · dbt · Apache Spark (distributed data processing)
- **Orchestration:** Apache Airflow / MWAA · Databricks Workflows · dbt · dependency-aware DAGs, SLA alerting, backfills, retries
- **Data Quality & Observability:** dbt tests · DLT expectations · Great Expectations · data reconciliation · Unity Catalog lineage
- **CI/CD for Data:** Databricks Asset Bundles · GitHub Actions · Terraform (IaC)
- **Cloud:** AWS (S3, Glue, MWAA, Lambda, IAM, Redshift) · Azure & Microsoft Fabric (OneLake, Direct Lake)
- **Data Modeling:** dimensional (star schema) · canonical / medallion modeling · data contracts
- **Focus areas:** data pipeline architecture · distributed data processing · orchestration · data quality & observability · CI/CD for data · stakeholder collaboration · technical leadership & mentorship

Professional Experience

Moody's — Assistant Director, Data Engineer (Staff-level IC) · Jan 2022 – Present · Bangalore, India

- Own the Orbis data product's ingestion and transformation pipelines end-to-end — designing fault-tolerant PySpark jobs on Databricks that conform 40+ heterogeneous global sources into a single trusted schema, processing **15+ TB/day**.
- Led the migration of **7 legacy PySpark pipelines** (~5,300 lines of wheel-packaged, DAB-deployed code) onto a standardized dbt + PySpark stack orchestrated by Airflow — with automated new-vs-legacy value-parity checks — **cutting hand-maintained pipeline code ~35%** and consolidating bespoke jobs into one repeatable, tested pattern.
- Built the team's data-quality and observability framework — automated validation gates (dbt tests, DLT expectations, Great Expectations; **85+ automated tests across the migrated reporting models alone**), quarantine flows, and end-to-end Unity Catalog lineage — **reducing data incidents and manual reconciliation effort 45%**.
- Cut Databricks compute spend **~35%** across environments by profiling workloads and tuning autoscaling policies, while sustaining delivery SLAs.
- Standardized CI/CD for data with Databricks Asset Bundles and GitHub Actions, scaling reusable ingestion and transformation patterns across **12+ production pipelines** and giving the team a consistent dev ' prod path.
- Made the design call to separate runtime-schema-driven jobs (PySpark, 2 pipelines) from config-driven models (dbt, 5 pipelines) — a classification standard the team now reuses to keep transformation logic testable and maintainable.
- Improved cross-team data usability by publishing data-ready events (SNS) and self-serve datasets consumed by **~4 downstream analytics and data-science teams**, backed by Unity Catalog table documentation and dbt docs (**200+ documented columns**); extended the Gold layer to Microsoft Fabric (Direct Lake) for near-real-time BI without duplication.
- Provide technical direction and mentorship to **6 engineers**, reviewing **~8–10 pull requests a week** to uphold data-modeling and pipeline standards; delivered the team's first GenAI proof-of-concept (RAG on Databricks Mosaic AI Vector Search) as a reference pattern for future AI/ML work.

FactSet Research Systems — Senior Data Engineer (Product Specialist II) · Feb 2019 – Jan 2022 · Hyderabad, India

- Built high-throughput market-data pipelines on Apache Spark and AWS (EC2, S3, Redshift), provisioned via Terraform, delivering reliable pricing data to client-facing products.
- Designed and shipped a data-quality web app using Isolation Forest for yield-curve outlier detection on spot-rate data, catching pricing anomalies before client delivery and **reducing downstream error escapes 50%**.
- Automated analytics workflows over FactSet CalcAPI / Formula API (OAS template generation, MBS performance reporting), **eliminating ~20 hours/week** of manual effort and turning ad-hoc product requests into self-serve pipelines.
- Partnered with the product team to build repeatable pipelines for usage data, turning ad-hoc requests into self-serve deliverables and reducing delivery turnaround time.

Franklin Templeton Investments — Data Engineer (Research Analyst) · Feb 2017 – Feb 2019 · Hyderabad, India

- Built ETL frameworks and data-warehousing solutions powering investment analytics and portfolio-management systems.
- Designed dimensional data marts (star schema) serving research analysts and portfolio managers with consistent, point-in-time datasets.
- Automated recurring research workflows in Python / SQL, improving data freshness and freeing analyst time for higher-value work.

S&P Global (via Genpact) — Data Engineer (Data Researcher II) · Oct 2014 – Jan 2017 · Hyderabad, India

- Built Python / SQL pipelines aggregating and normalizing multi-vendor pricing for US structured-finance securities (ISF / MBS) into a single source for the pricing desk.
- Automated structured-products operations (Console / PCDB workflows, FINRA / Intex feeds), **improving processing turnaround time by 30%**.
- Established early data-quality and reconciliation checks that improved pricing accuracy and downstream trust.

Education

B-Tech, Computer Science & Engineering — JNTU Anantapur (2007–2011) · MBA, Finance — Sri Venkateswara University (2012–2014)